Hypothesis Testing

Hypothesis Testing is a form of statistical inference that uses data from a sample to draw conclusions about a population parameter or a population probability distribution.

Hypothesis testing is an act in statistics whereby an analyst tests an assumption regarding a population parameter. The methodology employed by the analyst depends on the nature of the data used & the reason for the analysis.

* Key steps in Hypothesis Testing

1) State Hypotheses - Define the Null (H_0, no effect/difference) and Alternate (H_1, there is an effect/difference)

2) Set Significance Level (α) - choose the risk of error (commonly 0.05 or 5%)

3) Collect & Analyse Data - Gather sample data & compute a Test statistic (eg. Z, t) & its P-value

4) Make a Decision -

If P-value $< \alpha$ - Reject H_0 (evidence supports H_1)

If P-value $\geq \alpha$ - Fail to reject H_0 (not enough evidence against H_1)

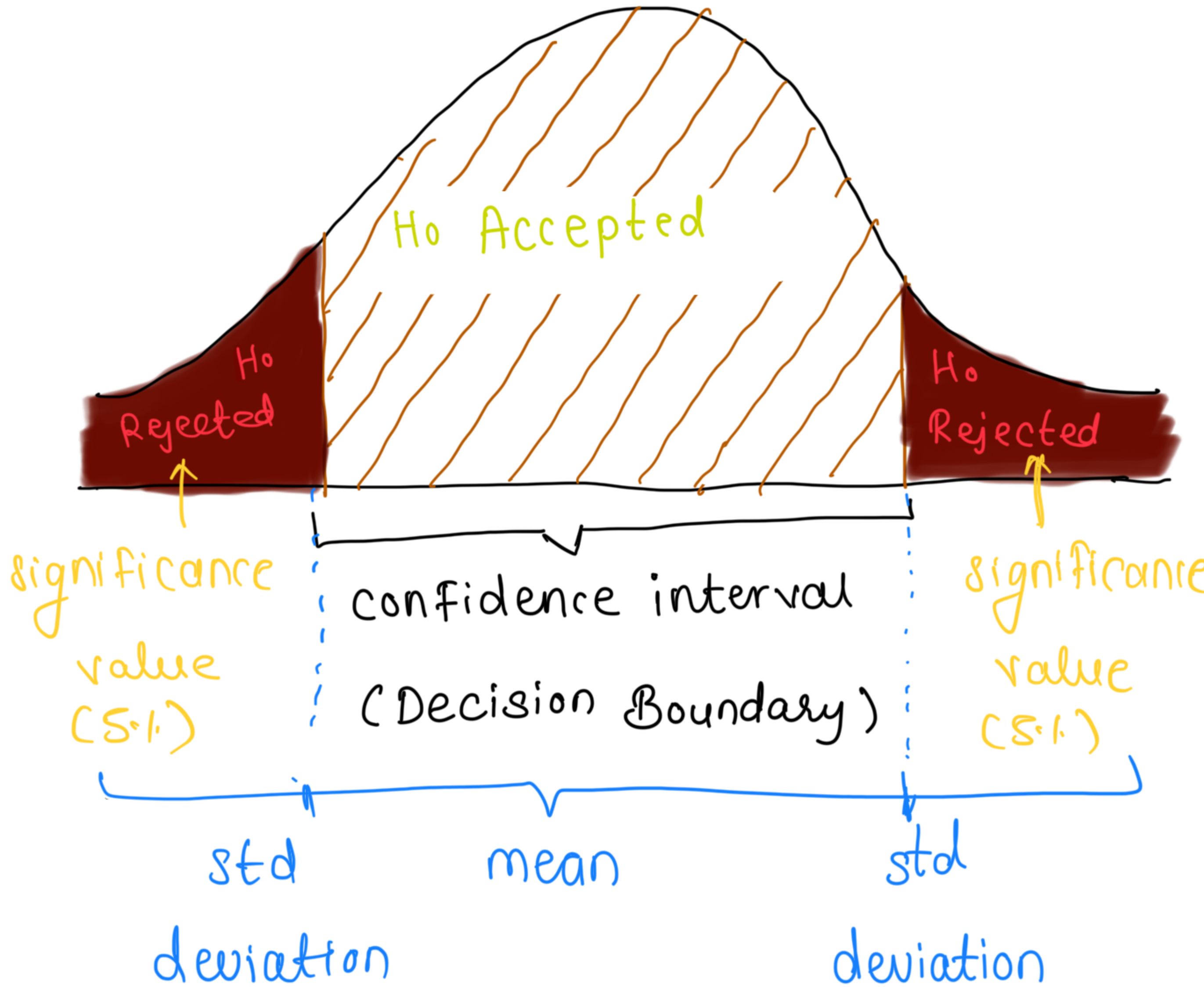
Core Concepts

Null Hypothesis (H_0) - The default assumption, often no stating no change or no relationship (eg a new drug has no effect)

Alternate Hypothesis (H_1) What you are trying to Find evidence for (eg the drug does lower blood pressure)

P-Value - Probability of observing your data (or more extreme) if the null hypothesis were true

Test Statistics - A value calculated from sample data (eg. Z-score, t-score) used to test the hypothesis.



Hypothesis Testing

Null Hypothesis (H_0)

Alternate Hypothesis (H_1)